

Day 11 – JavaScript: Events and DOM Manipulation

Objective:

Learn how to handle **user interactions (events)** and change or control the webpage using **DOM (Document Object Model) Manipulation**.

1. What is the DOM?

The **DOM (Document Object Model)** is a programming interface for HTML. It allows JavaScript to **access and change the content, structure, and style** of your webpage.

Example:

```
<p id="myPara">Hello!</p>
```

JavaScript can access this paragraph using:

```
let para = document.getElementById("myPara");  
para.innerText = "Hi, I changed!";
```

2. Accessing HTML Elements

Method	Use
<code>getElementById()</code>	Select element by ID
<code>getElementsByClassName()</code>	Select elements by class
<code>getElementsByTagName()</code>	Select elements by tag
<code>querySelector()</code>	Select using CSS selector

Example:

```
let title = document.querySelector("h1");  
title.style.color = "blue";
```

3. JavaScript Events

Events are actions that happen on a webpage (like clicks, hover, keypress, etc.)
You can write code that **runs when an event occurs**.

◆ Common Events:

Event	Triggered When...
onclick	Element is clicked
onmouseover	Mouse hovers over
onkeyup	Key is released
onsubmit	Form is submitted

4. Event Listener (Recommended Way)

```
document.getElementById("btn").addEventListener("click", function() {  
    alert("Button was clicked!");  
});
```

Example with DOM Manipulation:

```
<button id="btn">Click Me</button>  
<p id="text">Original Text</p>  
  
<script>  
    let btn = document.getElementById("btn");  
    let text = document.getElementById("text");  
  
    btn.addEventListener("click", function() {  
        text.innerText = "Text changed!";  
        text.style.color = "green";  
    });  
</script>
```

Practice Task:

1. Create a file: `day11.html`
 2. Add a heading `<h2>` and a button
 3. On button click:
 - Change the heading text
 - Change the color of the text
 4. Create a form with a name input field and submit button
 5. On form submit:
 - Prevent page reload using `event.preventDefault()`
 - Show alert with name input value
 6. Add mouseover event on a paragraph to change its background
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★ Submission Instructions:

- ✉ Send your file to: workdinarventure@gmail.com
- 📧 Subject: Day 11 - Events and DOM Manipulation
- ! Use this code as reference only
- 🔍 Google and experiment for better results
- ☐ Make your interaction smooth and user-friendly